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Honors Data Science

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Streets After Dark: Uncovering Crime Patterns In The Nations Capitol

1. Objective

This project looks at crime incident data from Washington, D.C. in 2024 to find patterns in crime types, locations, timing, and methods. It explores questions such as which crimes are most common, which wards have the most incidents, when crimes happen most often, what methods are used, and how crime volume changes during the year. By answering these questions, the analysis provides a clearer picture of public safety trends in the city. It is important to note that this analysis examines raw incident counts, not population-adjusted crime rates, meaning comparisons between wards reflect total reported incidents and may be influenced by differences in population size and reporting rates.

2. Introduction

The Crime Incidents dataset comes from DC Open Data (opendata.dc.gov), managed by the District of Columbia government. It contains 29,303 reported crime incidents from 2024 and includes 25 variables, such as offense type, ward, time shift, method used, block location, and report date.

Studying crime patterns in Washington, D.C. is important for several reasons. DC is a busy, diverse city with big differences between its eight wards, and crime data shows where and when people are most at risk. This dataset is helpful because it connects location, time, and crime type, giving a deeper look at public safety. Analyzing this data can reveal insights that help residents, policymakers, and city planners.

3. Method

I used Python and Jupyter Notebook to do the analysis. The main libraries were pandas for handling data, numpy for calculations, and matplotlib and seaborn for making charts.

First, I loaded the dataset with `pd.read_csv()` and checked it using `.head()`, `.dtypes`, `.shape`, and `.isnull().sum()` to see its structure, column types, and missing data. The dataset has 29,303 rows and 25 columns, with key variables like `OFFENSE`, `WARD`, `SHIFT`, `METHOD`, and `REPORT_DAT`.

To clean the data, I removed duplicate rows with `.drop_duplicates()`. Then I dropped rows with missing values in key columns like `OFFENSE`, `WARD`, `SHIFT`, `METHOD`, and `REPORT_DAT` using `.dropna()`. After cleaning, 29,299 rows were left, showing that very little data was lost and the dataset was already high quality.

Next, I converted the `REPORT_DATA` column from a string to a datetime object using `pd.to_datetime()`. I also made three new columns: `MONTH` (number), `DAY_OF_WEEK` (name), and `HOUR` (hour). These changes made it possible to analyze the data by time.

4. Storytelling: Data Visualization and Interpretation

Figure 1 – Crime Incidents by Type: The first chart is a horizontal bar graph that shows how often each crime type appears in the data. `THEFT/OTHER` is the most common, with over 12,000 incidents, nearly twice as many as the next most common, `THEFT F/AUTO` (auto theft), which has about 6,000. `MOTOR VEHICLE THEFT`, `ROBBERY`, and `ASSAULT W/DANGEROUS WEAPON` follow. Violent crimes like `HOMICIDE`, `SEX ABUSE`, and `ARSON` are much less frequent and are at the bottom of the chart.

This pattern shows that most crime in DC is property crime, not violent crime. The high number of thefts suggests that things like parked cars and easy access to personal belongings are major reasons for crime in the city. While violent crimes are still serious, their lower numbers mean that most crime in DC is not violent.

Figure 2 – Crime Incidents by Ward: The second chart is a bar graph comparing the total number of crimes in each of DC's eight wards. Ward 2 has the most incidents, with Ward 1 close behind. Ward 3 has the fewest crimes by a large margin, and Wards 5, 6, and 7 are in the middle.

Ward 2 includes areas like Downtown DC, Dupont Circle, and Georgetown, which are busy commercial and tourist spots with large crowds. This likely explains why crime is higher there, since more people and activity increase the chances of theft. Ward 3 is predominantly residential, which may contribute to fewer reported incidents, though this cannot be interpreted as a definitively safer ward without controlling for population size and reporting behavior.

Figure 3 – Crime Incidents by Time Shift: The third chart is a pie chart that shows crime incidents by time shift: Day, Evening, and Midnight. Evening makes up 40.5% of incidents, day is close behind at 39.3%, and midnight accounts for only 20.3%.

This result is a bit surprising because many people think crime is highest at night. But since most DC crime is property theft, it makes sense that crimes happen more during the day and evening when people are out. Fewer crimes at midnight do not mean nights are safe; it just shows that theft usually happens when people and their belongings are around.

Figure 4 – Crime Incidents by Method Used: The fourth chart is a bar graph showing the method used in crimes: OTHERS, GUN, and KNIFE. Most crimes, over 25,000, are in the OTHERS category, meaning no weapon was used. Gun-related crimes are in the low thousands, and knife crimes are rare. This matches the earlier finding that most DC crime is property theft, which usually does not involve weapons. While gun crimes are less common, they are still a real safety concern, especially in cases like robbery and assault with a dangerous weapon. Knife crimes are the least common, so when violence does happen, it is more likely to involve guns than knives in DC.

Figure 5 – Monthly Crime Trends: The fifth chart is a line graph showing crime numbers for each month in 2024. Crime is lower in January and February, rises in the spring, and peaks in July and August with over 2,800 incidents each month. It then drops through the fall and into December.

This seasonal trend is common in criminology research and is often linked to more outdoor activity during warmer months. When more people are outside, there are more chances for theft, assault, and other crimes. In winter, fewer people are outside, so there are fewer crimes. This suggests the city could add more police and resources in the summer to handle the increase in crime.

Figure 6 – Offense Type by Ward: A grouped bar chart comparing offense types across wards shows that THEFT/OTHER dominates in nearly every ward but is especially concentrated in Wards 1, 2, and 5. Motor vehicle theft is notably higher in Wards 5 and 7. This reveals that the type of crime varies meaningfully by location, not just the total count.

Figure 7 – Violent Crime by Time Shift: Violent crimes including robbery, assault, homicide, and sex abuse — are more concentrated in the Evening and Midnight shifts compared to property crimes. This suggests different policing strategies may be needed depending on crime type and time of day.

5. Conclusion: The most important finding is that DC crime is mostly property-based: THEFT/OTHER is the main offense type, most crimes involve no weapon, and most incidents happen during the day and evening when people and their property are most accessible. Looking at locations, Ward 2 has the most crime, likely because it has many businesses and tourists. Ward 3, which is more residential, has the lowest incident count.. Over time, crime peaks in summer and drops in winter, a pattern seen in other crime studies too. These findings have practical uses. Policymakers and law enforcement can use this data to better plan where and when to send patrols, like adding more officers in busy wards during evenings and summer, and focusing on theft prevention in crowded areas. Future studies could look at crime type by ward and time to find even more specific patterns. Overall, data science is a valuable tool for understanding and improving city safety.

6. References

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